

PROFESSIONAL SUMMARY

Data Scientist and ML Engineer with research and production experience in machine learning, LLM systems, and MLOps. Experienced deploying distributed ML pipelines and GPU-accelerated LLM serving on Kubernetes, building agentic AI applications, and developing novel ML methods published in peer-reviewed venues. Strong foundation in statistical modeling, deep learning, and time-series forecasting.

EDUCATION

- Master of Science in Data Science** Berlin, Germany  
*Berlin University of Applied Sciences (BHT) — Graduated With Distinction (1.06 German)* Oct. 2024 – Sept. 2026
- Bachelor of Science in Computer Science; Minors in Math & Stats** Bellingham, WA  
*Western Washington University — GPA: 3.96* Sept. 2020 – June 2024

EXPERIENCE

- Data Science Researcher (Machine Learning & Engineering)** Berlin, Germany  
*Calgo Lab — Berlin University of Applied Sciences (BHT)* Feb. 2025 – Sept. 2026
  - Engineered distributed ML experiment pipelines on a Kubernetes cluster with Helm; automated testing and linting via GitHub Actions CI/CD.
  - Deployed and maintained Dockerized LLM serving stack (vLLM on Nvidia GPUs) and production web-demo applications.
  - Architected **MechDetect**, a novel ML framework for classifying error mechanisms in tabular datasets; published at IEEE DSIS 2025.
  - Implemented spatiotemporal groundwater-level forecasting (master’s thesis) combining Chronos2, N-HiTS, and GPyTorch Gaussian Processes (Kronecker product, Matérn) across 499 monitoring sites.
- Computer Science Researcher (Deep Learning)** Bellingham, WA  
*Hutch Research — Western Washington University* Mar. 2023 – June 2024
  - Built a PyTorch/TensorFlow probabilistic neural network pipeline to predict binary star system parameters; published in *The Astronomical Journal*, Vol. 171 (2026).
  - Optimized large astronomical dataset processing with NumPy; maintained code quality and reproducibility in a 4-person team.
- Mathematics Researcher (Statistical Analysis)** Bellingham, WA  
*Western Washington University* Oct. 2022 – June 2024
  - Developed a novel class of non-parametric two-sample tests in R; presented at Joint Mathematics Meetings 2024 and SIAM PNW 2023.

SELECTED PROJECTS

- LangGraph Email Agent** | *vLLM, LangGraph, Kubernetes, Docker, Nvidia A100*
  - Deployed an agentic LLM email-answering system on Kubernetes using vLLM for inference serving and LangGraph for multi-step orchestration with structured output.
- Tax Law RAG System** | *BERT Fine-tuning, LangChain, FAISS, Kubernetes*
  - Built a production RAG pipeline with fine-tuned BERT retrieval; benchmarked FAISS vs. Annoy vector databases on English and German corpora. Published the fine-tuned BERT to HuggingFace.
- Berlin PM2.5 Spatiotemporal Modeling** | *Chronos2, Kubernetes, Docker, Streamlit*
  - Automated daily ETL and inference pipeline on Kubernetes; spatiotemporal forecasting via Chronos2 foundation model and ordinary kriging.
- tab-err: Tabular Data Error Library** | *Python, PyPI, CI/CD, GitHub Actions*
  - Contributed to open-source library for realistic tabular error generation; developed high-level APIs and comprehensive unit tests.

PUBLICATIONS

- Anonymized but Useful Synthetic Tabular Health Data for AI based Fall Risk Assessment** | *Scientific Data* 2026
- MechDetect: Detecting Data-Dependent Errors** | *IEEE DSIS* 2026
- Probabilistic Neural Networks for Eclipsing Binary Parameters** | *The Astronomical Journal* 2026
- Towards Realistic Error Models for Tabular Data** | *ACM JDIQ* 2025

SKILLS

**MLOps & Deployment:** Kubernetes, Docker, Helm, vLLM, FastAPI, GitHub Actions, Prometheus, Grafana  
**LLM & Agentic AI:** LangChain, LangGraph, FAISS, RAG, Prompt Engineering, Structured Outputs, Agentic Pipelines, Transformer Fine-tuning  
**ML & Data:** PyTorch, TensorFlow, GPyTorch, Scikit-learn, Pandas, NumPy, PostgreSQL, MySQL, Time-Series, Spatial Data  
**Cloud & DevOps:** AWS, GPU Workloads (Nvidia A100), Containerization, Git, Linux  
**Languages:** Python, SQL, R, Go, Bash, Java   **Spoken:** English (Native), German (B1)